Reg. No. | | | | | | | | | | |

[17CIC42]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

AUTOMATA THEORY AND LOGIC

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full questions.Choosing at least **TWO** questions from each part.
2. Any missing data may be suitably assumed.

PART - A

1. a. Define substring, prefix of a string, suffix of a string and Language with example. **04**

b. i. Design a DFA to accept strings of c and d not having substring cdd. Show the moves for string cdcdc. **06**

ii. Design a DFA to accept strings of 0's and 1's such that number of 0's is multiple of 2 and number of 1's multiple 3.

iii. Design a DFA to accepts strings of 0's and 1's such that every block of 5 symbols have at least two 1's.

c. Design a DFA to accept binary strings divisible 5 and should start with 1. **04**

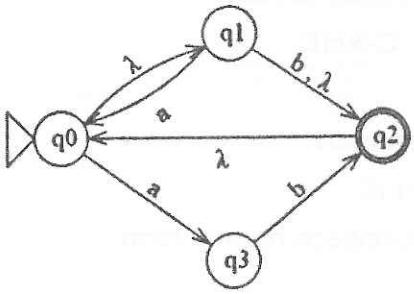
2. a. Design a NDFSM for the following language and convert it into DFSM. **05**

L={w={a,b}* : fourth symbol from the last character is a}

b. Define NDFSM and Language accepted by NDFSM. **03**

c. i. Define epsilon closure. **02**

ii. Convert the following NDFSM to DFSM. **02**



3. a. Define regular expression. **02**

b. Write regular expression for the following languages **08**

L={W:W={a,b}* W has aab as substring}

L={W:W={a,b}* |w| mod 3=0}

L={anbm: n<=3 m>=4}

L={ w | w contains at least two a's, or exactly two b's }.

c. Prove that for every regular expression r there is a NFA N such that L(r)=L(N). **04**

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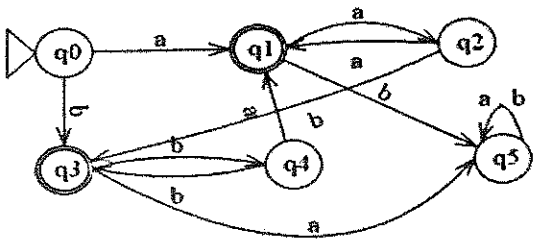
4. a. Let M be the following DFSM. Use minDFSM to minimize M. **08**



b. Prove that following language is not regular **06**

L={0P: P is prime number}

L={W: W has equal number of a's and b's

PART - B

5. a. i. Define Grammar and ambiguous grammar **03**

ii. Design a grammar for the following languages **06**

L={anbn : n>0}

L={W: W={a,b}* |W| mod 4=0}

L={WWR| W={a,b}*}

b. Prove that following grammar is ambiguous and write unambiguous grammar. **05**

S->iCtS | iCtSeS|a C->b

6. a. Define PDA, language accepted by final states and empty stack. **03**

b. Design a PDA for the following language **06**

L={ambn : m≠n,m,n>0}. Show the moves made by PDA for the string aaaabbb

c. Convert the following grammar to PDA. **05**

S-> AB | BC A->a B->b C->c|Ɛ

7. a. Simplify the following grammar. **08**

S->Aa|B B->a|bC C->a|Ɛ

b. Convert the following to Greibach normal form **06**

S → XA|BB

B → b|SB

X → b

A → a

8. a. Explain Multitape Turing machine and non-deterministic Turing machine. **06**

b. Design a Turing Machine to compute the following function. **08**

f(x,y)=x-y x>y

= 0 otherwise

Draw transition diagram. Show the moves for the string 0010000